

UMM E FARWA

Civil Engineer

📍 Karachi, Pakistan

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SUMMARY

B.E. Civil Engineer with a **CGPA 3.55/4.0**, specializing in structural engineering. Actively engaged in research on ultra-high-performance concrete (UHPC) and the application of artificial intelligence, particularly artificial neural networks (ANN), for advanced construction materials. Strong interest in structural systems, structural mechanics, and innovative data-driven approaches in Civil Engineering. Intends to pursue a Master's degree focused on intelligent and resilient infrastructure systems.

EDUCATION

Bachelor of Engineering in Civil Engineering – CGPA 3.55 / 4.0 <i>NED University of Engineering & Technology Karachi</i> • Key Coursework: Structural Analysis I & II, Reinforced Concrete Design I & II, Numerical Analysis, Soil Mechanics I & II, Transportation I & II and Construction Project Management.	2022-26
Intermediate (HSSC) in Pre-Engineering – 85.45% – Grade A-1 <i>St. Lawrence's Govt. Girls Degree College Karachi</i>	2020-22
Matriculation (SSC) in Science – 86.24% – Grade A-1 <i>Jinnah Foundation Secondary School Karachi</i>	2018-20

INTERNSHIPS EXPERIENCE

Paragon Constructor Private Limited (EMAAR Project) <i>Site Engineer Internee – Karachi, Pakistan – 1 month</i> • Conducted daily site inspections to monitor work quality, ensure adherence to safety regulations, and verify compliance with approved structural drawings and specifications. • Assisted in foundational activities, including layout, boring, and concreting for pile foundations, emphasizing precision and quality control. • Prepared detailed Reinforced Concrete (RC) element schedules and Bar Bending Schedules (BBS) for beams, columns, and slabs.	2025
AAA Partnership Private Limited <i>BIM Modeler Internee – Karachi, Pakistan – 1 month</i> • Modelled a 400 sq. yard residential house in Autodesk Revit, achieving accurate 3D architectural and structural representations. • Applied Building Information Modelling (BIM) standards for model organization and coordinated between CAD drawings and the Revit model to ensure design integrity. • Completed training in project management using Procore, gaining experience in document control and project communication	2025

TECHNICAL & COMPUTATIONAL SKILLS

• Structural / FEA: ETABS, structural analysis, moment distribution, finite element method	(Skilled)
• BIM & Drafting: Autodesk Revit, Autodesk AutoCAD, reinforcement detailing	(Advanced)
• Programming / AI: Python, MATLAB, ANN-based modeling, computational logic	(Intermediate)
• Microsoft Office: MS PowerPoint, MS Word, MS Excel, MS Project	(Advanced)
• Project Management: Procore training, work breakdown structures (WBS), construction costing	(Beginner)
• Laboratory Skills: CTM/UTM operation, concrete mix design, aggregate testing, tensile testing	(Beginner)

RESEARCH INTEREST

- **Computational & Structural Mechanics:** Constitutive modeling of advanced materials, Finite Element Analysis (FEA), structural analysis and design, and complex non-linear material behavior.
- **Artificial Intelligence (AI):** Application of Artificial Neural Networks (ANN) and Machine Learning (ML) for predictive modeling, structural performance assessment, and data-driven design and optimization.
- **Advanced & Sustainable Materials:** High and Ultra-High-Strength Concrete (HSC/UHSC), advanced cementitious materials, sustainable construction materials, and their applications in high-performance structural systems.

PROJECTS

Constitutive Modelling of Advanced Concrete using AI Neural Networks 2025-26

Final Year Design Project – NED University

- **Scope:** Focused on developing a high-fidelity, data-driven constitutive model to predict the complex non-linear behavior of **High and Ultra-High-Strength Concrete (HSC/UHSC)**.
- **Methodology:** Leveraged **Artificial Neural Networks (ANN)** and advanced computational tools to process experimental material data, moving beyond traditional empirical constitutive laws.
- **Impact:** The **ANN-based model** is constructed to estimate the key mechanical parameters of UHPC, supporting experimental analysis and optimization of mix design.

Structural Analysis & Design of Academic Block N1 2024

Academic Project – NED University

- Designed the main structural elements (beams, columns, slabs, footings) for an Academic Block using **manual calculations, Moment Distribution Method, Finite Element Approach**, and validated the results using ETABS.
- Produced detailed structural drawings in **AutoCAD** and 3D models in **Revit** for comprehensive representation and detailing.

Python Based Structural Analysis Tool Development 2025

Academic Project – NED University

- **Developed a Python-based application with a Graphical User Interface (GUI)** for the design and analysis of continuous beams in accordance with **ACI-318** standards.
- Applied computational logic to automate tedious design checks, demonstrating proficiency in both structural code and programming logic.

Experimental and Laboratory Work 2022-25

Materials Testing Laboratory – NED University

- Executed **concrete mix design** and performed essential material quality control tests, including slump test, water absorption, and aggregate impact value.
- Conducted **compression testing** on cubes and beams (CTM/UTM) and performed **tensile strength testing of steel** (UTM) to gather data for material performance evaluation.

HONORS & CERTIFICATIONS

- **Certification (2026):** Ultra High Strength Concrete - From Brittle to Impact Resistant.
- **Prime Minister's Youth Laptop Scheme (2025):** Awarded Laptop based on academic performance.
- **Certification (2024):** Completed training in project management software **PROCORE**.
- **Competition (2024):** Participated in the **ACI BAMBOOZLE** concrete beam design competition.
- **Contest:** Participated in **International Kangaroo Linguistic Contest**.

CONFERENCES & SEMINARS

- **Seminar Attendance:** Attended a seminar on the impact of **Artificial Intelligence** in aviation, with a focus on computational applications.
- Attended a seminar on **Geo-Synthetic Case Studies** held at the NED University Auditorium.

MEMBERSHIPS

- Member of the American Society of Civil Engineers (ASCE) 2026
- Member of the American Concrete Institute (ACI) 2026
- Member of the Structural Engineering Institute of ASCE 2026